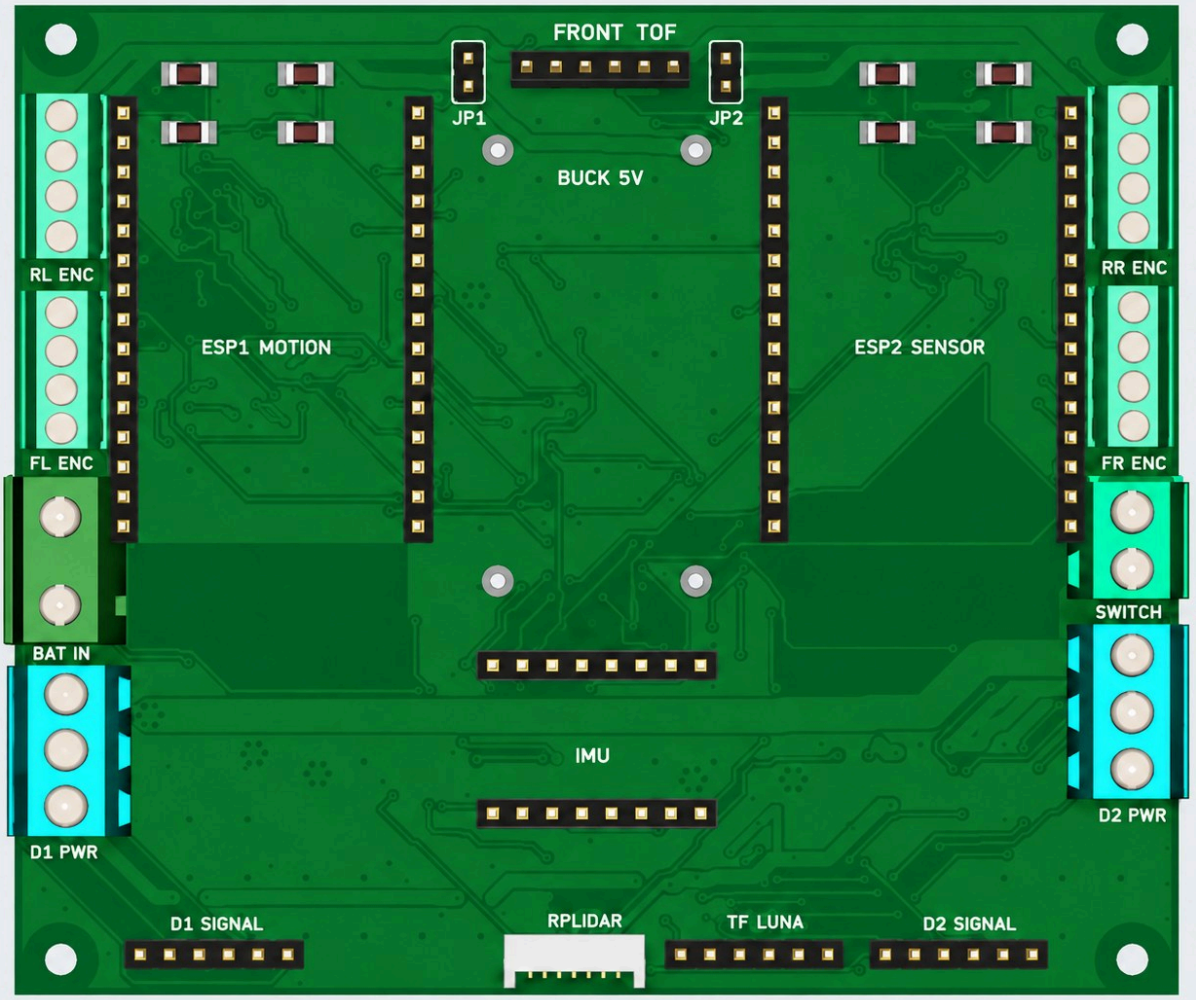
Smart Car Carrier Board

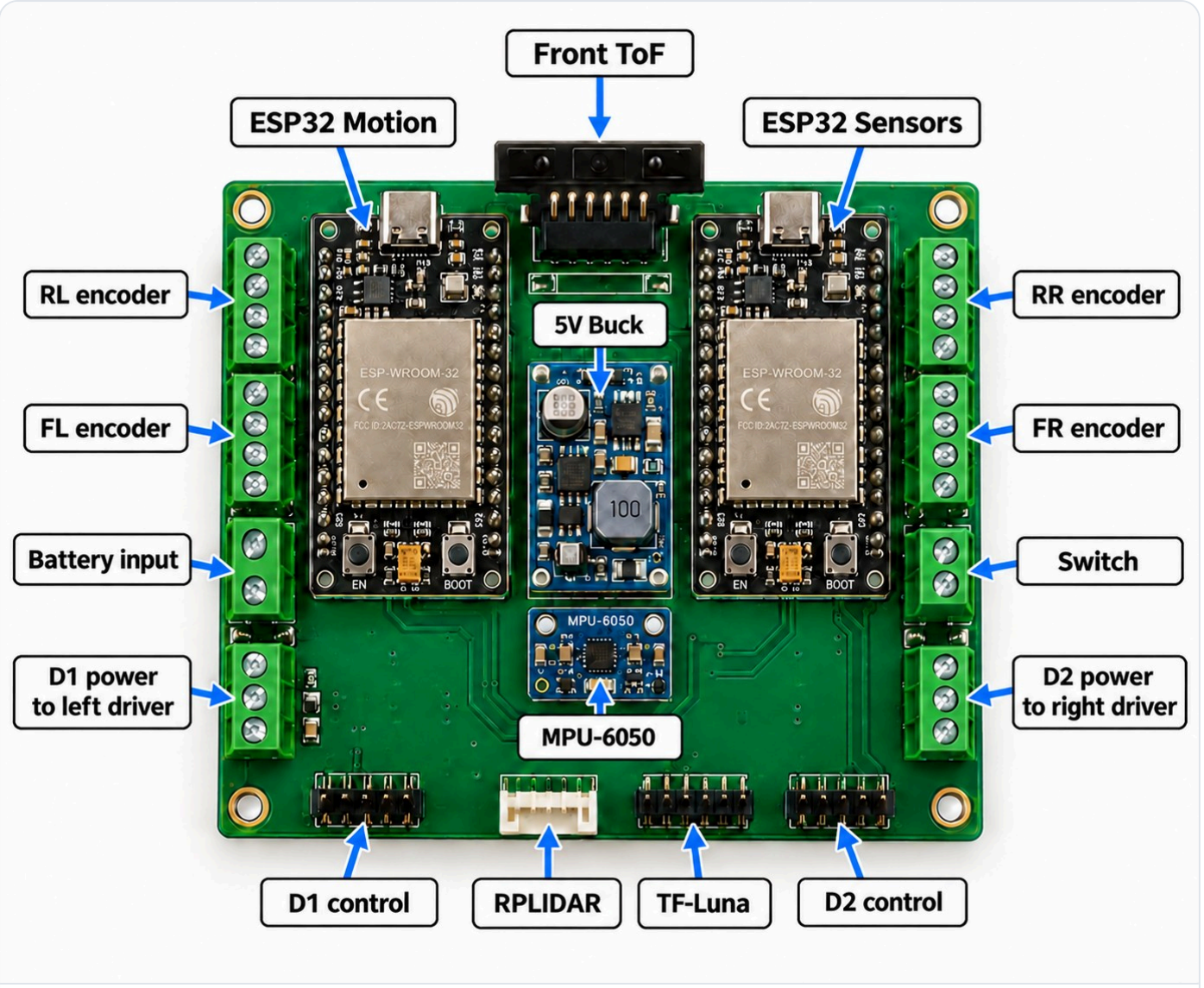
Assembly and wiring guide · 100 × 85 mm · all views component side up, FRONT TOF at the top

1 Bare board



Top side. Board labels match the names used throughout this guide.

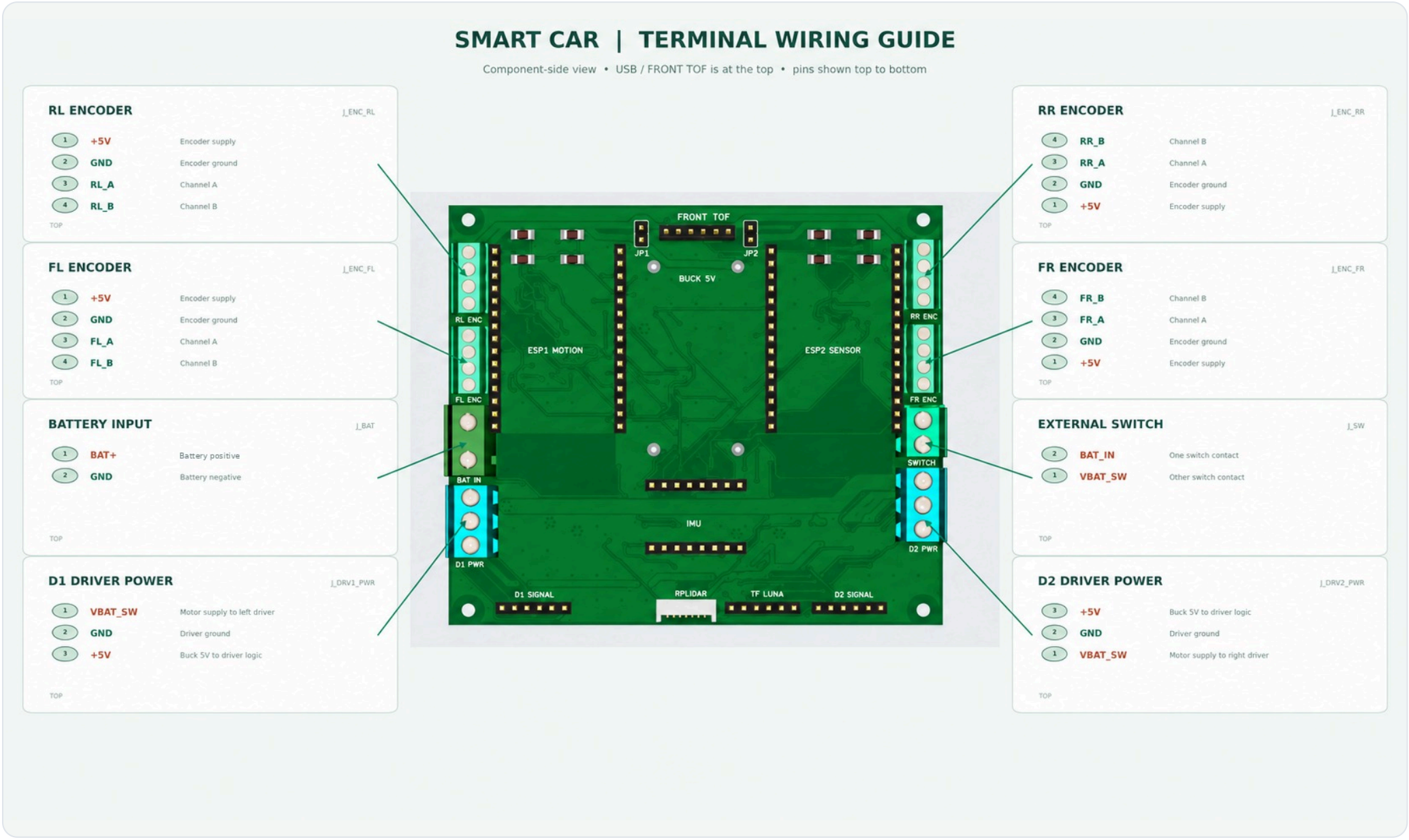
2 Module placement



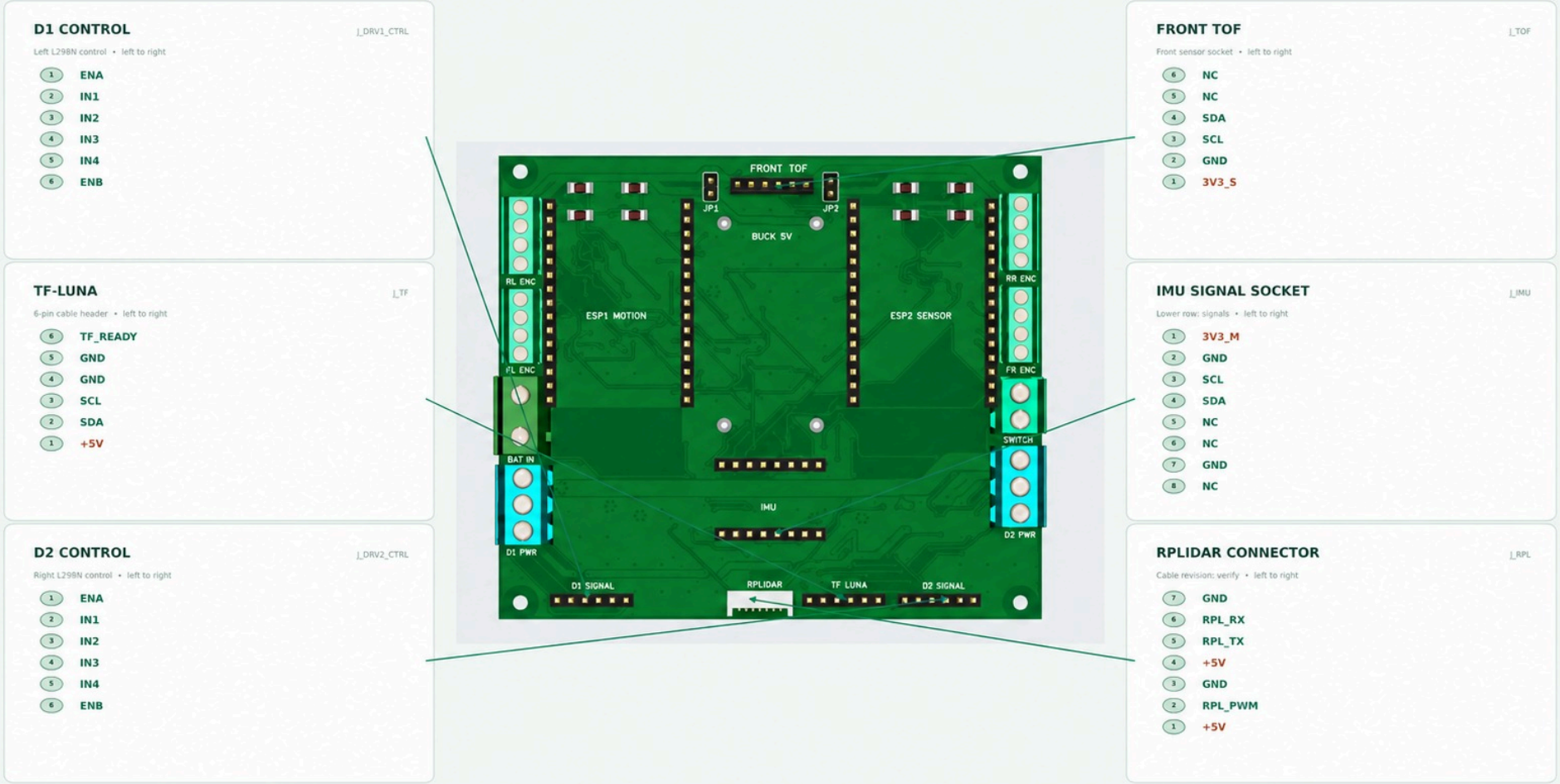
Where each module mounts on the carrier.

BOARD LABEL	FIT	MOUNTING
ESP1 MOTION	ESP32 DevKit 30-pin	Into 2 × 15 female rows, USB toward FRONT TOF
ESP2 SENSOR	ESP32 DevKit 30-pin	Into 2 × 15 female rows, USB toward FRONT TOF
BUCK 5V	LM2596HVS module	Soldered on 4 pads · IN toward IMU, OUT toward FRONT TOF
IMU	GY-521 (MPU-6050)	Lower row = signals · upper row = mechanical support
FRONT TOF	VL53L0X	Upright at front edge, lens facing forward
RPLIDAR	RPLiDAR A1M8	7-pin 1.25 mm cable
TF LUNA	TF-Luna	6-wire cable, female Dupont ends
D1 / D2 SIGNAL	Left / right L298N	6 jumper wires each
D1 / D2 PWR	Left / right L298N	3-pole screw terminal
FL · RL · FR · RR ENC	600 PPR encoders	4-pole screw terminal, ferruled wires
BAT IN · SWITCH	3S pack · panel switch	2-pole screw terminals

3 Screw terminals: pin order

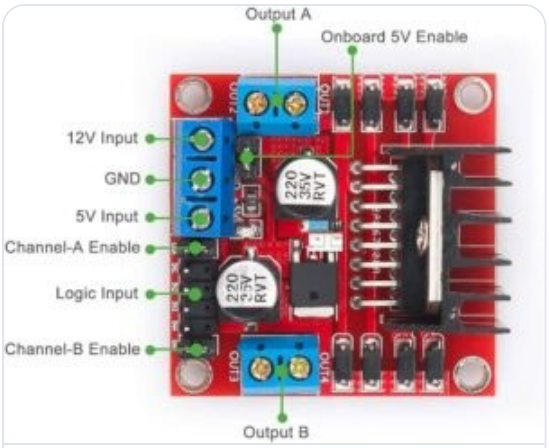


4 Pin headers: pin order



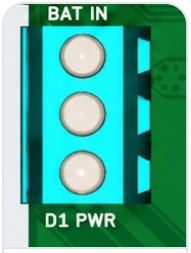
5 L298N motor drivers

MODULE



L298N terminal and pin names.

ON THE BOARD



D1 PWR (left edge)



D1 SIGNAL



D2 PWR (right edge)



D2 SIGNAL

Power terminals

D1 = LEFT driver · D2 = RIGHT driver

D1 PWR · top → bottom

BOARD	DRIVER
Top	12V Input
Middle	GND
Bottom	5V Input

D2 PWR · top → bottom

BOARD	DRIVER
Top	5V Input
Middle	GND
Bottom	12V Input

Signal headers

same order on both

D1 SIGNAL → D1 · D2 SIGNAL → D2 · left → right

BOARD PIN	DRIVER
1	Channel-A Enable (ENA)
2	Logic Input IN1
3	Logic Input IN2
4	Logic Input IN3
5	Logic Input IN4
6	Channel-B Enable (ENB)

Motors

DRIVER	MOTOR
D1 Output A · B	Front-left · Rear-left
D2 Output A · B	Front-right · Rear-right

6 Encoders, battery and switch

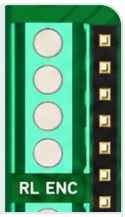
MODULE



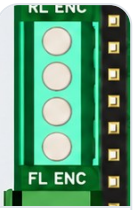
600 P/R

600 PPR incremental encoder, 4 wires.

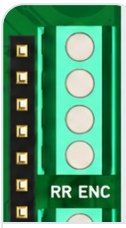
ON THE BOARD



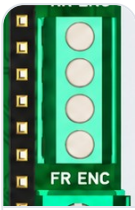
RL



FL



RR



FR

Encoders

FL · RL · FR · RR ENC

Left side: FL ENC, RL ENC · top → bottom

BOARD	WIRE
+5V	Red
GND	Black
A	Green
B	White

Right side: FR ENC, RR ENC · top → bottom

BOARD	WIRE
B	White
A	Green
GND	Black
+5V	Red



XT60 battery connector



BAT IN (left edge)

Battery

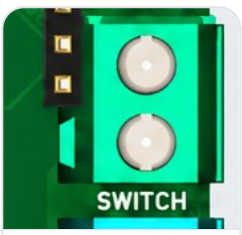
BAT IN

Top → bottom · short pigtail to an XT60

BOARD	XT60
Top · BAT+	+ (positive)
Bottom · GND	–



KCD4 panel rocker switch



SWITCH (right edge)

Switch

SWITCH

Top → bottom · switch mounted in the enclosure

BOARD	SWITCH
Top · BAT_IN	Contact 1
Bottom · VBAT_SW	Contact 2 (same pole)

7 5V buck and TF-Luna



LM2596HVS buck module



BUCK 5V area (centre)

5V buck

BUCK 5V

MODULE PAD	BOARD NET
IN+	VBAT_SW (battery after switch)
IN−	GND
OUT+	+5V rail
OUT−	GND

Set to 5.0 V before connecting any load

1. Power the buck input only, 5V load disconnected.
2. Multimeter on DC volts across OUT+ / OUT−.
3. Turn the trim screw until it reads 5.0 V.
4. Power off, then connect boards and sensors.

MODULE



TF-Luna with its 6-wire cable.

CONFIGURATION



Sensor connector pins 1 → 6.

ON THE BOARD



TF LUNA header (bottom edge)

TF-Luna

TF LUNA

Board header, left → right

BOARD	TF-LUNA PIN
1 · TF_READY	6 · Data ready
2 · GND	5 · Configuration input (GND = I2C)
3 · GND	4 · GND
4 · SCL	3 · TXD / SCL
5 · SDA	2 · RXD / SDA
6 · +5V	1 · +5V

8 RPLiDAR and front ToF

MODULE



RPLiDAR A1M8 with interface board and cable.

ON THE BOARD



RPLIDAR connector (bottom edge)

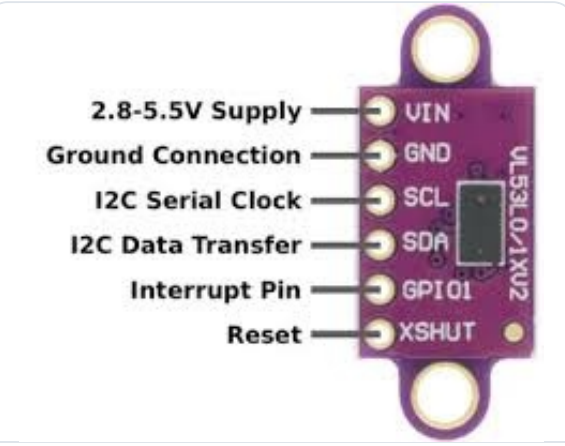
RPLiDAR A1M8

RPLIDAR · 7-pin 1.25 mm

Board connector, left → right

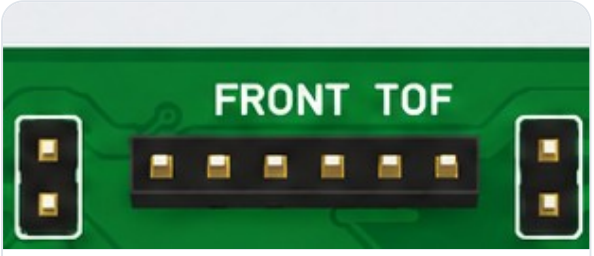
BOARD	LIDAR
1 · GND	GND
2 · RPL_RX	RX
3 · RPL_TX	TX
4 · +5V	VCC_5 (core supply)
5 · GND	GND_MOTO
6 · RPL_PWM	MOTOCTL
7 · +5V	5V_MOTO (motor supply)

MODULE



VL53L0X pin names.

ON THE BOARD



FRONT TOF header (top edge)

VL53L0X ToF

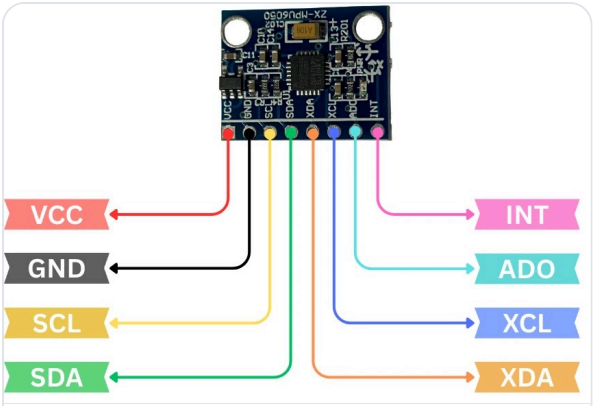
FRONT TOF

Board header, left → right

BOARD	SENSOR
1 · NC	XSHUT
2 · NC	GPIO1
3 · SDA	SDA
4 · SCL	SCL
5 · GND	GND
6 · 3.3V	VIN

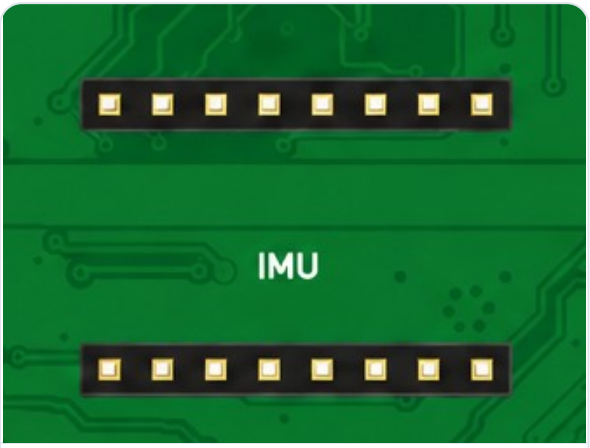
9 IMU

MODULE



MPU-6050 pin names.

ON THE BOARD



IMU socket: lower row = signals, upper row = support

MPU-6050 IMU

IMU · lower row

Lower header row, left → right

BOARD	GY-521
1 · 3.3V	VCC
2 · GND	GND
3 · SCL	SCL
4 · SDA	SDA
5 · NC	XDA (not used)
6 · NC	XCL (not used)
7 · GND	AD0 (address 0x68)
8 · NC	INT (not used)